

Syllabus and Study guide for AI Final exam

Sources

- Russell & Norvig's textbook (AIMA)
- Slides available in the AIMA book website
- Slides and handnotes posted via Google classroom

Study guide

- The exam is **cumulative**, i.e., everything taught in this course will be covered in the final exam. But I have trimmed some topics, as you will see in the table.

Syllabus

Ch.	Subject	Topics/Skills required
1	Intro to AI	• Understanding the basics of AI, and knowing its applications in different domains • Understanding the four basic approaches to AI (Think humanly, think rationally, etc.)
2	Intelligent Agents	• Understanding PEAS description • Understanding the characteristics of environments (static/dynamic, stochastic/deterministic, etc.) • Understanding different types of models (model-based, utility based, etc.)
3	Search	• Understanding all blind and heuristic search techniques (BFS, DFS, Greedy, A* etc.) and ability to solve mathematical problems related to these.
5	Adversarial Search	• Understand minimax and alpha-beta pruning and ability to solve simple mathematical problems related to these. • Understand techniques to make game tree search faster (cut-off search, move ordering, etc.)
6	CSP	• Understand CSP basics, backtracking, forward checking, heuristics to improve backtracking, arc consistency. • Use these to solve CSP problems

7,8	Logic	• Translating English sentences to and from propositional logic and first order logic • Converting sentences to and from CNF and Horn form • Applying resolution, forward and backward chaining on propositional logic statements
13	Quantifying Uncertainty	• Understanding basic notions of probability, conditional probability, independence, full joint distribution, Bayes' rule
14	Probabilistic Reasoning	• Understanding conditional independence • Understanding the basic principle of Bayesian Networks • Ability to make inference given the topology and conditional probability tables of a given Bayesian network
15	Temporal Reasoning	• Understanding the basic principles of a Markov process • Ability to perform basic calculations given the transition table of a Markov process • Understanding the basic principles of a Hidden Markov Model (HMM) • Ability to make inference (filtering) given the transition and emission probabilities of a Hidden Markov Model
18	Machine Learning Basics	• Understanding the basics of four major types of machine learning (supervised, unsupervised, semi-supervised, reinforcement) • Understanding concepts related to supervised learning: training, testing, learning model, overfitting, underfitting • Understanding basic working principle of perceptron and neural networks